

Suhani Shukla

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Education

University of California — Los Angeles (expected June 2027)

B.S. Computer Science, Minor in Entrepreneurship | GPA: 3.7/4.0

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Networking, Machine Learning, Deep Learning, NLP

Honors and Awards: Dean's Honors List, National Merit Scholar (2023), UCLA QWER Hacks 3rd Place (2024), AIME Qualifier (2021)

Experience

Machine Learning Software Engineer Intern (Applied AI/ML), Microsoft AI

June - Sept. 2026

- Developed Small Language Models (SLMs) for query fanout powering Microsoft WebIQ. Built end-to-end Python ML experimentation pipelines spanning data generation, curation, supervised fine-tuning, and **reinforcement learning**.
- Benchmarked and implemented SLM serving optimizations, including KV caching and quantization, evaluating inference latency, model quality, and serving cost tradeoffs for production deployment at scale.
- Developed a Gemma-270M regression-based trigger model for dynamic query fanout, predicting search-quality gains to balance fanout utility and API call savings; identified predictive signals for routing decisions across experiments.
- Improved a critical internal search-quality metric +**2.43** points (**3.3%**) over baseline across post-training techniques, with resulting models selected for production. Reduced search engine API calls by **24%**, improving search efficiency.

Software Development Engineer Intern, Amazon Web Services

June - Sept. 2025

- Automated end-to-end CI/CD workflows for AWS Redshift Console deployments using Live Pipeline Templates (Ruby) and AWS CDK (TypeScript), migrating legacy infrastructure to easily reproducible Infrastructure as Code (IaC).
- Onboarded services onto Apollo containers and Application Revisions for deterministic builds and standardized releases across environments. Designed a zero-touch rollback system with real-time canary analysis and composite alarms, minimizing blast radius for faulty deployments and ensuring safe global releases across **20+ AWS regions**.
- Reduced deployment cycle time from **100+ to under 28 days** and increased deployment frequency **4x** through staged bake times, automated approvals, and promotion blockers. Integrated automated testing, performance monitoring, and deployment safety checks into release pipelines, improving reliability for thousands of enterprise customers globally.

AI Consultant & Technical Lead Bruin AI

Sept. 2023 - Present

- Led delivery of a **\$30K** client engagement, Bruin AI's largest to date, built an AI-native marketing and CRM platform for the Art Agency. Consolidated thousands of records onto AWS RDS, with LLM- and OCR-powered creator-brand matching and metric-driven campaign guidance; translated client feedback into shipped product improvements.
- Built and deployed a production RAG-based HR assistant grounded in internal company documents for AlphaCore Wealth Advisory using FastAPI, FAISS vector search, embeddings, AWS Lambda, and S3, now serving **100+** employees.

Co-President & Founding Member, AWS Cloud Club at UCLA

Sept. 2025 - Present

- Co-founded UCLA's first cloud computing club and scaled the organization from 6 to 30+ active members. Led cloud and AI workshops and organized Architect the Cloud, an event with 90+ attendees and 10 industry speakers.
- Established project partnerships with **AWS** and **UCLA Digital Technology Solutions** within the club's second quarter; oversaw delivery, interfaced with stakeholders, and translated client requirements into technical solutions.

Software Engineering Intern, Nokia

June - Aug. 2024

- Developed diagnostic software in C++ for ASIC components, improving reliability and observability of workflows.
 - Implemented bidirectional gRPC streaming for inter-process communication in C++ for custom CLI (Diags), enabling real-time system-wide logging and diagnostics across test suites. Ensured robustness for prolonged ASIC runs.
 - Designed and implemented custom POSIX signal handler to safely manage interrupts via Ctrl-C for long-running diagnostics. Integrated with Python front-end for improved usability for test engineers and safer orchestration.
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Projects

EMG-to-QWERTY Keystroke Prediction (Deep Learning Project) 

- Evaluated CNN, LSTM/GRU, Transformer, and CNN-Transformer hybrid models for EMG-based keystroke prediction.
- Applied data augmentation and analyzed overfitting tradeoffs; benchmarked models using Character Error Rate.

Easy EBT: Grocery Store EBT Item Finder 

- Built full-stack application to find and purchase EBT/SNAP-eligible items at grocery stores using Python web scraping and recipe recommendations, with React JS and Django. Awarded third place at UCLA QWER Hacks.
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Skills

Languages: C++, C, Python, TypeScript, JavaScript, Bash

AI/ML: PyTorch, HuggingFace, LangChain, LLMs, Small Language Models (SLMs), Reinforcement Learning, Supervised Fine-Tuning, RAG, Embeddings, CNNs, Model Evaluation, Amazon Bedrock, Azure Machine Learning

Web & Infrastructure: React, HTML/CSS, FastAPI, Django, MongoDB, Tailwind CSS/Material UI, Docker, CI/CD, Infra as Code